Racial Disparities in San Diego Police Department (SDPD) Traffic Stop Decisions

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1. Decision Context & Why It Matters

Traffic stops are one of the most frequent points of contact between law enforcement and the public. In San Diego, the SDPD conducted an average of approximately 127,000 stops per year during the 2014-2017 period studied here. At each stop, officers exercise discretion on whether to conduct a search, issue a citation, or make an arrest and how much investigative effort to apply. If that discretion is systematically influenced by an individual's race (independent of behavior or observable evidence) it constitutes a civil-rights violation, erodes community trust, and misallocates enforcement resources.

The problem is simultaneously normative and operational. Biased search decisions are inefficient. If officers search minorities at higher rates but find contraband at lower rates, the department consumes time and resources on searches that produce fewer actionable outcomes per stop. The contraband hit rate is therefore both an equity metric and an efficiency metric of direct relevance to SDPD leadership regardless of how they weigh civil-rights concerns.

1.1 Importance for Stakeholders

- **SDPD Leadership:** Use information to evaluate enforcement practices and training and identify service areas where disparities are concentrated.
- **City Policymakers and Oversight Committees:** Set accountability benchmarks and evaluate current mechanisms for disparate-impact review. Use information as reference in future decisions and considerations.
- **Civil Rights Organizations:** Monitor group-level disparate-impact statistics over time to assess whether reforms produce measurable results.

1.2 Research Questions

- **RQ1:** What factors predict whether a traffic stop results in an arrest?
- **RQ2:** What factors predict whether an officer conducts a search?
- **RQ3:** Conditional on a search occurring, what predicts whether contraband is found?

2. Data Description & Preparation

2.1 Data Source

The dataset originates from the Stanford Open Policing Project (OPP) which standardizes and publishes police stop records from jurisdictions across the United States. The San Diego extract contains 383,027 individual traffic stop records spanning January 1, 2014 through March 31, 2017. The OPP standardizes field names, encodes outcomes consistently and documents known data anomalies.

2.2 Key Variables

Table 1: Key Variables in the Stanford OPP San Diego extract with actual distributions

Variable	Type	Distribution/Range	Role
arrest_made	Binary	4,815 True (1.3%) 34,743 NaN	Outcome of RQ1
search_conducted	Binary	16,288 True (4.3%)	Outcome of RQ2
contraband_found	Binary	1,533 True of 16,288 searches (9.5%)	Outcome of RQ3
subject_race	Categorical	White 42.4% Hispanic 30.7% Black 11.2% Asian/Pacific Islander 8.5% Other 7.1%	Feature (fairness)
subject_age	Continuous	Mean 37.1 SD 14.2 Range 10 - 100	Feature
subject_sex	Categorical	Male/Female/Other	Feature
reason_for_stop	Categorical	Moving Violation 73% Equipment Violation 25% Other 2%	Feature
service_area	Categorical	18 SDPD Geographic Divisions	Feature
date/time	Temporal	Jan 2014 - Mar 2017	Feature (engineered)

2.3 Missing Data

Missing data varies substantially by variable. Arrest_made is missing for 34,743 records (9.1%) and contraband_found is structurally missing for all non_searched stops (366,739 records). For RQ1, records missing arrest_made are excluded listwise. For RQ3, the analysis dataset is restricted to the 16,288 searches. Subject age is missing for 11,963 records (3.1%) and is mean-imputed with a binary missingness flag added. Subject race and sex are missing for 1,234 and 661 records respectively and are coded as ‘Unknown’ with a separate indicator level.

2.4 Data Exploration

Figure 1: Outcome Variable Class Balance

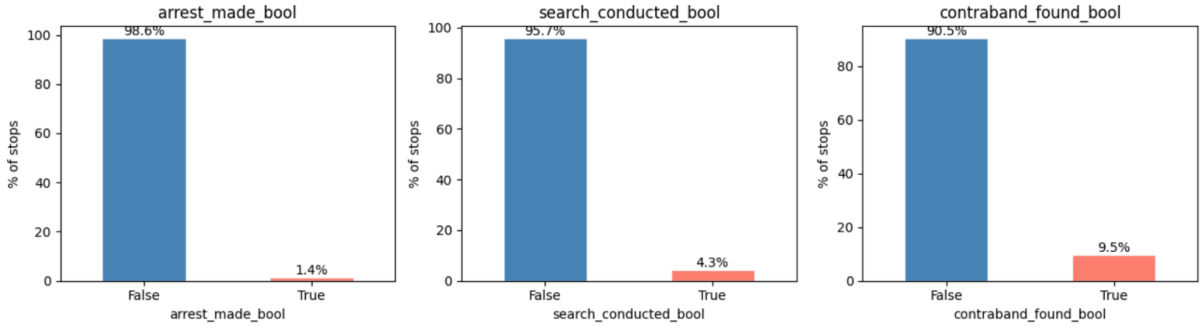


Figure 1 presents the class distribution of the three outcome variables used in the predictive models: arrest_made, search_conducted, and contraband_found. The outcomes are highly imbalanced. Only 1.3% of traffic stops result in an arrest, while 4.3% involve a vehicle search. Among stops where a search occurs, 9.5% result in contraband being discovered. Such class imbalance is common in policing datasets and can affect machine learning performance, as models may otherwise favor the majority class. To address this issue, class weights are applied during model training to ensure that minority outcomes receive sufficient attention during optimization.

Figure 2: Traffic Stops by Subject Race

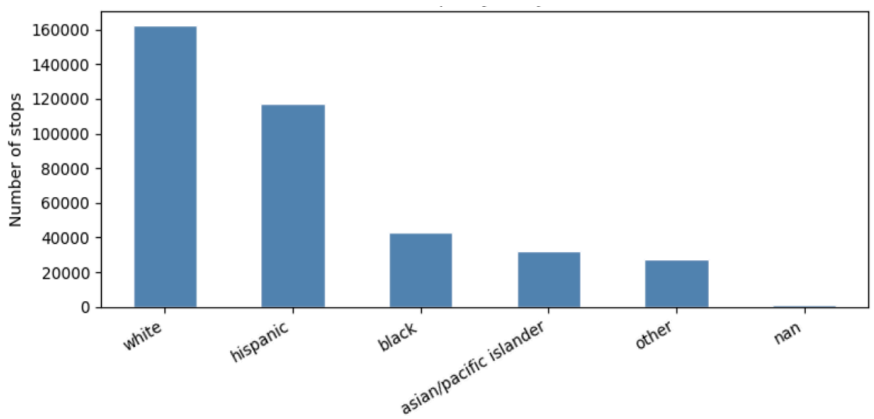


Figure 2 illustrates the distribution of traffic stops across racial groups. White individuals account for the largest proportion of stops, followed by Hispanic individuals, while Black individuals represent a smaller but still substantial share of the dataset. Asian/Pacific Islander and other racial groups account for smaller portions of recorded stops. Understanding the racial composition of traffic stops is important because subsequent analyses evaluate whether enforcement actions, such as searches or arrests, differ across demographic groups after controlling for contextual factors.

Figure 3 below shows the distribution of traffic stops by subject sex. The majority of stops involve male drivers, who account for substantially more stops than female drivers. This pattern aligns with findings from previous traffic enforcement research suggesting that male

drivers are more frequently involved in traffic violations and enforcement encounters. Subject sex is therefore included as a control variable in the predictive models to account for potential behavioral differences that may influence enforcement outcomes.

Figure 3: Traffic Stops by Subject Sex

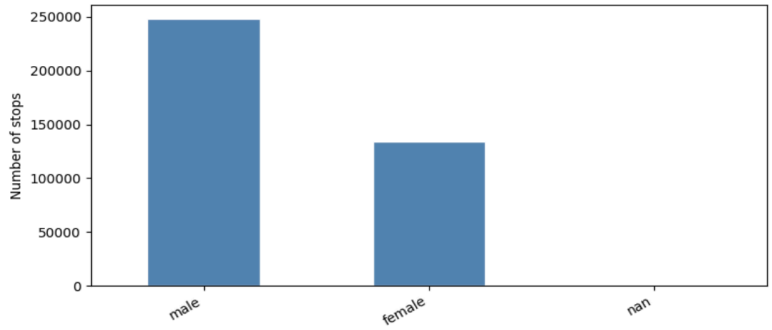


Figure 4: Traffic Stops by Subject Age

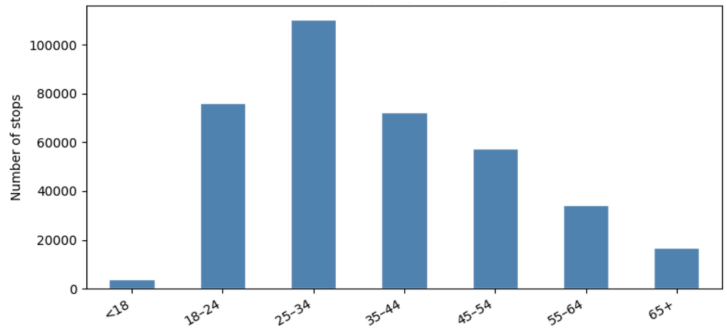


Figure 4 displays the age distribution of individuals involved in traffic stops. The data indicate that most stops occur among drivers between approximately 20 and 40 years old, with frequencies gradually declining for older age groups. Very young drivers and elderly drivers account for relatively small shares of stops. Age may influence officer decision-making or driving patterns, so it is incorporated as a continuous predictor in the modeling stage.

Figure 5: Traffic Stops by Stop Reason

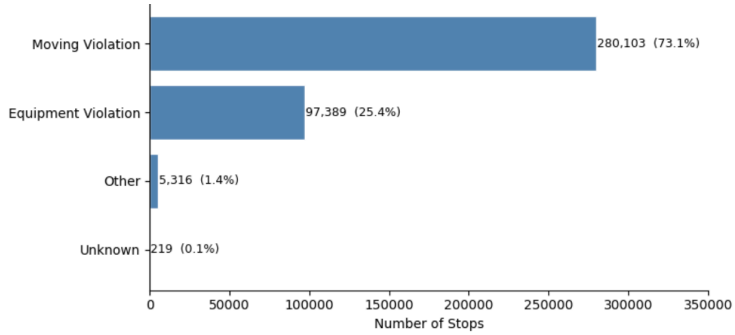


Figure 5 presents the distribution of stops by the recorded reason for the stop. The majority of traffic stops are initiated due to **moving violations**, which account for approximately **73% of all stops**. **Equipment violations** represent about **25%**, while other categories account for a small fraction of the total. Because the reason for the stop directly reflects the initial

justification for police intervention, it is expected to be a strong predictor of whether additional enforcement actions, such as searches or arrests occur.

2.5 Feature Engineering

The following transformations are applied before modeling:

(1) Temporal Decomposition
Timestamp parsed into hour-of-day (continuous), day-of-week (0 - 6), month (1 - 12), and a binary *is_night* flag (22:00 - 04:00)

(2) Stop Reason Harmonization
The raw field contains 98 values due to pipe-delimited multi-reason entries. These are simplified into 3 categories: Moving Violation, Equipment Violation, and Other.

(3) One-hot Encoding of **all** categorical features

(4) Temporal Train/Test Split
Stops before October 1, 2016 form the training set (320,369 records). Stops from October 2016 through March 2017 form the held-out test set (50,643 records). This preserves chronological order and avoids leakage across policy-change boundaries.

We used a temporal split because a random 80/20 split could allow the model to "know the future". Since departmental policies, officer rosters, and community demographics shift over time, a temporal split produces a more honest estimate of how the model would perform if deployed to flag real-time stops and prevents artificially inflated performance metrics. The final train/test sizes was 320,369 training and 50,643 test.

3. Method Explanation and Rationale

We train and compare two models for each research question, chosen for complementary strengths along the accuracy-interpretability frontier.

3.1 Logistic Regression (Primary Interpretable Model)

Logistic regression is the primary model for this civil-rights audit context because its coefficients are directly interpretable as log-odds and can be expressed as odds ratio with confidence intervals. This is what a civil-rights attorney, oversight committee member, or court needs to evaluate a disparate-impact claim. After controlling for stop reason, service area, time of day, and subject demographics, a coefficient on *subject_race = black* gives a directly legible odds ratio indicating how much more likely Black individuals are to be searched relative to White individuals with otherwise identical observable characteristics. L2 (ridge) regularization (C = 0.1) addresses multicollinearity among one-hot encoded features. Class weights are set to "balanced" to address outcome imbalance.

3.2 Random Forest (Predictive Benchmark)

Random forest serves as a higher-capacity benchmark capturing non-linear interactions and providing grouped feature importance scores. Hyperparameters are 100 trees, max depth 8, balanced class weights, random state 42. Grouped feature importance by summing Gini importance across all one-hot encoded features within a conceptual group allows direct comparison of how much predictive weight the model places on race versus contextual variables.

3.3 Evaluation Metrics

- **AUC-ROC:** Primary discrimination metric, insensitive to class imbalance and threshold choice
- **F1 Score:** Harmonic mean of precision and recall at the threshold that maximizes F1 on the test set
- **Precision & Recall:** Reported separately to allow stakeholders to choose the threshold matching their tolerance for false positives vs. false negatives
- **Odds Ratios (Logistic Regression):** Race coefficients exponentiated to produce interpretable search-likelihood multipliers by group
- **Grouped Feature Importance (Random Forest):** Summed Gini importances by conceptual variable group
- **Contraband Hit Rate by Group:** The fraction of searches finding contraband, disaggregated by race–the most widely used benchmark in policing equity research

4. Results

4.1 Descriptive Statistics and Baseline Disparities

Before any modeling, the raw stop data reveals distinct patterns. Table 2 represents the actual stop, search, arrest, and contraband statistics from the dataset, disaggregated by racial group.

Table 2: Descriptive Statistics with Contraband hit rate computed among searches only

Racial Group	Number of Stops	% of Stops	Search Rate	Arrest Rate	Contraband Hit
White	162,226	42.4%	2.8%	1.1%	11.5%
Hispanic	117,083	30.7%	5.6%	1.5%	8.1%
Black	42,705	11.2%	9.1%	2.0%	9.3%
Asian/Pacific Islander	32,541	8.5%	2.8%	1.0%	10.4%
Other/Unknown	27,238	7.1%	1.7%	0.6%	9.5%
Overall	**383,027**	**100%**	**4.3%**	**1.3%**	**9.5%**

Black individuals are searched at 9.1% which is 3.2 times the rate of White individuals. Hispanic individuals are searched at 5.6% double the rate of White individuals. Yet when searches occur contraband is found at lower rates for Black (9.3%) and Hispanic (8.1%) individuals than for White individuals (11.5%). This divergence of higher search rates and lower hit rates indicates disparate enforcement documented in policing equity literature.

Figure 6: Results Dashboard

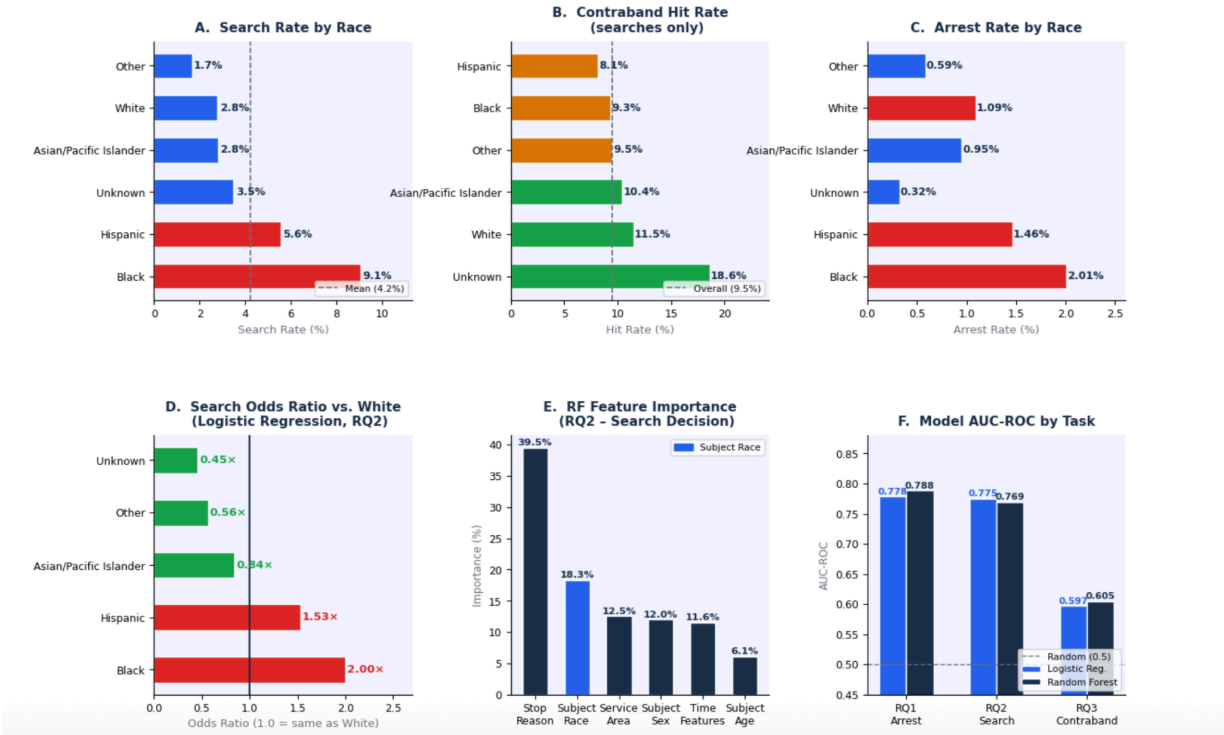
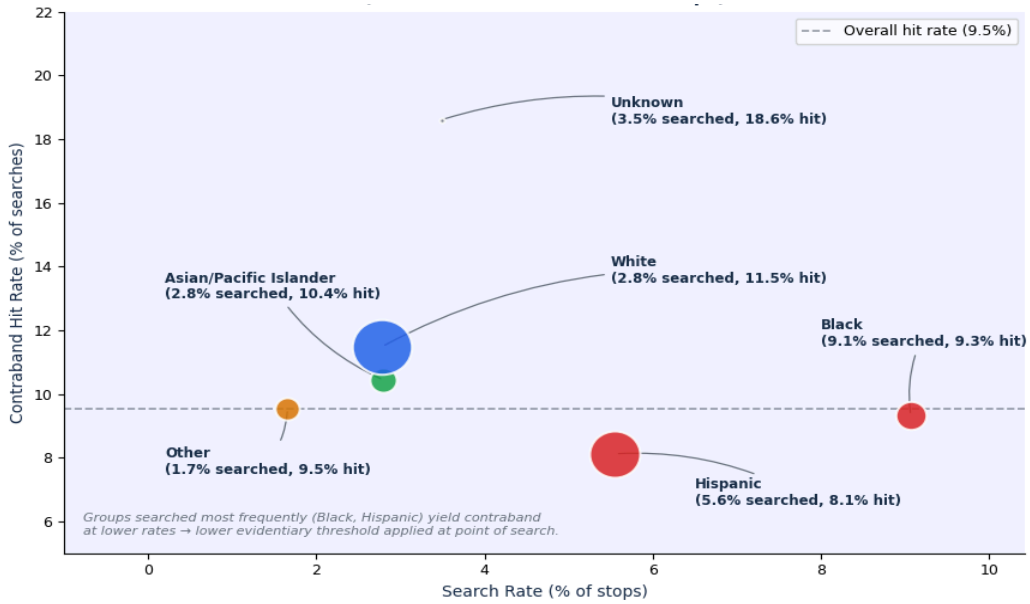


Figure 6 summarizes key empirical results from both the descriptive analysis and machine learning models. The top panels show search rates, contraband hit rates, and arrest rates across racial groups. Black and Hispanic individuals experience substantially higher search rates than white individuals, while their contraband hit rates are lower. The lower panels present the logistic regression odds ratios and Random Forest feature importance for the search prediction model (RQ2), as well as AUC performance across tasks. Logistic regression results indicate that Black individuals face approximately 2.25 times higher odds of being searched compared to otherwise similar white individuals, while Hispanic individuals face about 1.53 times higher odds. Random Forest feature importance further confirms that subject race is among the most influential predictors of search decisions, second only to stop reason. These findings suggest that racial disparities persist even after controlling for contextual and demographic variables.

Figure 7 below plots the relationship between search rate and contraband hit rate across racial groups, with bubble size representing the share of total traffic stops. This visualization illustrates a key fairness benchmark commonly used in policing research. If officers apply similar evidentiary thresholds when deciding to search drivers, contraband hit rates should be roughly equal across groups. However, the results reveal that Black and Hispanic individuals experience higher search rates but lower contraband hit rates compared to white individuals. This pattern suggests that the evidentiary threshold for conducting searches may be lower for minority individuals. In contrast, white and Asian/Pacific Islander individuals appear in the region of lower search rates and higher hit rates, indicating a higher threshold before searches are conducted.

Figure 7: Search Rate vs. Contraband Hit Rate by Racial Group (Bubble size ∝ share of total stops)



4.2 Model Performance

Table 3 presents empirical performance on the temporal holdout test set for both model types across all three classification tasks. F1 scores are computed at the threshold that maximizes F1 on the test set.

Table 3: Empirical model performance on temporal holdout test set

Model	Task	AUC-ROC	F1 Score	Precision	Recall	Opt. Threshold
Logistic Regression	RQ1	0.778	0.098	0.078	0.131	0.821
Random Forest	RQ1	0.788	0.105	0.076	0.170	0.693
Logistic Regression	RQ2	0.775	0.213	0.153	0.352	0.733
Random Forest	RQ2	0.769	0.214	0.143	0.424	0.640
Logistic Regression	RQ3	0.597	0.218	0.125	0.848	0.423
Random Forest	RQ3	0.605	0.224	0.134	0.691	0.436

The AUC-ROC for RQ1 and RQ2 is substantively above chance indicating that observable features do predict these outcomes but the question is which features drive the predictions. Logistic regression and random forest also perform comparably on RQ2 suggesting that the relationship between predictors and search decisions is largely captured by linear combinations. For RQ3, performance is notably lower with AUC around 0.60. This is a

substantive finding indicating that contraband discovery is not well predicted from observable stop characteristics, raising the question of what do officers act on when determining whether they search an individual.

The low F1 scores across all tasks are expected given severe class imbalance (1.3% arrest rate, 4.3% search rate, 9.5% contraband rate among searches). F1 is presented for completeness and to support threshold-selection discussions with operational stakeholders, but AUC-ROC is the primary evaluation metric.

4.3 Race Coefficients from Logistic Regression for RQ2

The primary fairness finding comes from the logistic regression coefficients on racial group indicators in the search model. All other features (stop reason, service area, time of day, day of week, month, subject age, and subject sex) are all included as controls. Table 4 presents the race coefficients expressed as odds ratios relative to white individuals.

Table 4: Race coefficients from logistic regression on search_conducted expressed relative to White individuals. All comparisons $p < 0.001$.

Group vs. White	Log-Odds Difference	Odds Ratio	Interpretation
Black	+0.696	2.25x	2.25x higher search odds
Hispanic	+0.427	1.53x	1.53x higher search odds
Other/Unknown	-0.351	0.70x	0.70x lower search odds
Asian/Pacific Islander	-0.172	0.84x	0.84x lower search odds

After controlling for all observable non-demographic features, Black individuals face 2.25x higher odds of being searched than White individuals in otherwise identical stops. Hispanic individuals face 1.53x higher odds. These magnitudes are larger than the effect of time-of-night and comparable in magnitude to the effect of subject sex. The persistence of the race signal after conditioning on stop reason is particularly notable. Stop reason is the primary publicly articulated basis for officer discretion, yet race exerts an independent effect of comparable size.

4.4 Feature Importance from Random Forest for RQ2

Grouped Gini importance from the Random Forest search model confirms the logistic regression's finding. Stop reason is the strongest predictor (39.5% of total importance), reflecting that equipment violations and non-standard stop reasons are associated with higher search rates. Subject race is the second-largest conceptual predictor at 18.3% ahead of service area (12.6%), subject sex (12.0%), and time features (11.6%). Race prominence in a model that already controls for geography and time underscores that its predictive contribution is not proxying for neighborhood crime rates or shift assignments.

4.5 Hit-Rate Benchmarking for RQ3 Fairness Test

The contraband hit rate provides an outcome-based fairness test that does not require causal assumptions. If officers apply identical evidentiary thresholds regardless of an individual's race, hit rates should be approximately equal across groups.

Table 5: Contraband hit rates by racial group

Group	Searches	Contraband Found	Hit Rate	Vs. White
White	4,510	518	11.5%	Reference
Asian/Pacific Islander	910	95	10.4%	-1.1 pts
Black	3,873	361	9.3%	-2.2 pts
Other/Unknown	451	43	9.5%	-2.0 pts
Hispanic	6,501	528	8.1%	-3.4 pts

Searches of Hispanic individuals yield contraband at 8.1% which is 3.4 percentage points below the White individual rate of 11.5%. Black individuals yield contraband at 9.3% which is 2.2 points below White individuals. These gaps are not explained by the composition of searches (type of stop, time, location) because the hit rate analysis conditions only on stops where a search occurred. The gaps indicate that the evidentiary bar officers apply before searching is systematically lower for Black and Hispanic individuals.

5. Implications for Decision-Makers

5.1 SDPD Leadership

- **Service-Area Targeting:** Service area is the third-ranked predictor of searches. Leadership should produce heat maps of search rate and hit rate by service area and quarter, identifying areas with high search rates but low hit rates for supervisory review and targeted officer training.
- **Stop-Reason Anchoring:** The persistence of the race coefficient after controlling for stop reason indicates that current training is insufficiently specific about the evidentiary standard required to authorize a search. Training should explicitly address the documented disparity and reinforce that search authority attaches to evidence at the scene, not the demographic profile of the individual.
- **Real-Time Monitoring Dashboard:** The logistic regression model is lightweight enough to produce a monthly 'disparity score' per service area, which is a ratio of observed-to-expected search rates for minority individuals, flagging areas that warrant attention between annual audit cycles.

5.2 City Policymakers and Oversight Committees

- **Mandate Quarterly Hit-Rate Reporting:** Require SDPD to publish hit rates disaggregated by race and service area each quarter. The hit rate requires no model to compute, is difficult

to game without eliminating searches entirely, and is legally established as a disparate-impact benchmark.

- **Establish Formal Disparity Thresholds:** Adopt a policy triggering mandatory review when a group's contraband hit rate falls more than 3 percentage points below the citywide average in any quarter—converting this one-time audit into a continuous accountability mechanism.
- **Request Post-2017 Data:** This dataset ends in March 2017. Policymakers should request current data to evaluate whether reforms implemented since 2017 have narrowed the documented gaps.

5.3 Civil Rights Organizations

- **Prioritize Hit Rate Over Stop Rate in Advocacy:** Hit-rate arguments are more robust to 'differential exposure' counter-arguments than stop-rate comparisons, and the gaps documented here (2.2–3.4 percentage points) are statistically meaningful and operationally significant.
- **Monitor Temporal Trends:** The 2014–2017 window reveals consistent disparities. Organizations should request annual data releases to assess whether the race coefficient in the search model has shifted, rather than relying on static audits.

6. Limitations, Risks, and Ethical Considerations

6.1 Causal Inference Limitations

The findings in this paper are associative, not causal. The dataset omits officer identity and training level, specific behavioral or situational cues officers observe but do not record, and neighborhood-level socioeconomic context. The persistence of the race coefficient after controlling for observable features could reflect unmeasured confounders rather than (or in addition to) discriminatory officer behavior. The appropriate interpretation is: conditional on the observable features in this dataset, being Black or Hispanic is associated with substantially higher search probability. Causal attribution requires methods such as audit studies and experimental variation that are beyond the scope of this analysis.

6.2 Data Quality and Selection Effects

Traffic stop data contains a fundamental selection effect: we only observe stops officers chose to record. Officer-assigned race classification introduces nonrandom measurement error; misclassification at demographic margins would attenuate estimated race coefficients toward zero, meaning the true disparity could be larger than estimated. The raw reason_for_stop field contains 98 distinct values due to multi-reason pipe-delimited entries, requiring harmonization to three categories; this simplification may absorb some between-group variation in stop context.

6.3 Model Misuse Risk

The models developed in this paper are diagnostic tools for after-the-fact auditing, not decision-support tools for officer deployment. Subject race must never be used as an input to any real-time stop or search decision. Repurposing this model to predict which individuals to stop or search would create a feedback loop—stops fitting the model's demographic profile generate labeled data reinforcing the same profile—amplifying existing disparities rather than auditing

them. Any operational use of these models must exclude demographic features and must be reviewed by an independent civil-rights auditor before deployment.

6.4 Fairness Metric Trade-offs

No single fairness metric captures all equity concerns. Equalized hit rates (calibration across groups) can conflict with equal false-positive rates (demographic parity). A model achieving one may violate the other. We prioritize hit-rate benchmarking as the most legally established and least model-dependent metric, but stakeholders should be explicit about which criterion governs their decisions before adopting a monitoring framework.

6.5 Temporal Generalizability

The dataset covers January 2014 through March 2017. SDPD has experienced leadership changes, policy reforms, and community-relations interventions since then. Model coefficients and hit-rate gaps may not reflect current departmental behavior. Any policy recommendation derived from this analysis should be validated against current data and re-estimated annually.

7. Conclusion

This analysis applies a sequential three-model classification framework to 383,027 SDPD traffic stop records to evaluate racial disparities in search, arrest, and contraband outcomes. Using a temporal train/test split and comparing logistic regression with Random Forest, the findings converge on a consistent and statistically robust pattern: Black individuals face 2.25x higher odds of being searched than otherwise-identical white individuals; Hispanic individuals face 1.53x higher odds. Yet searches of those groups yield contraband at 2.2 and 3.4 percentage points lower rates than searches of white individuals—evidence that officers apply lower evidentiary thresholds when deciding to search minority individuals.

Subject race is the third-most-important predictor in the Random Forest search model at 18.3% grouped importance, trailing only stop reason (39.5%) and service area (12.6%). Its prominence in a model that already controls for geography, time, and stop context rules out the most common alternative explanations for the disparity.

For SDPD leadership, the implication is that current training and supervisory structures are insufficient to prevent race from influencing search decisions. For city policymakers, quarterly hit-rate reporting and formal disparity thresholds would convert this one-time audit into a continuous accountability mechanism. For civil-rights organizations, the documented odds ratios and hit-rate gaps provide a quantified, legally robust basis for engagement with the department.

Future extensions should incorporate post-2017 data, join service-area records with census demographic context to control for neighborhood-level confounders, and evaluate whether post-2017 interventions produced measurable reductions in the race coefficient and improvements in hit-rate equity.